

Lab 13: The Sleuth Kit

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>

Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+

- The Sleuth Kit
- Assume we have the image file
 - <https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd>

Step 1.

- Download a USB image

```
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jun  5 10:41:56 2025 from 10.0.2.2
(kali㉿kali)-[~]
$ mkdir usb

(kali㉿kali)-[~]
$ cd usb

(kali㉿kali)-[~/usb]
$ wget https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
--2025-07-09 11:57:42-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
Resolving www.dropbox.com (www.dropbox.com)... 162.125.81.18, 2620:100:6031:18::a27d:51
12
```

```
(kali㉿kali)-[~/usb]
$ ls Ch01InChap01.dd -l
-rw-rw-r-- 1 kali kali 1474560 Jul  9 11:57 Ch01InChap01.dd

(kali㉿kali)-[~/usb]
$ md5um Ch01InChap01.dd
Command 'md5um' not found, did you mean:
  command 'mdsum' from deb ucommon-utils
  command 'md5sum' from deb coreutils
Try: sudo apt install <deb name>

(kali㉿kali)-[~/usb]
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbbcb  Ch01InChap01.dd
```

- Tool will show the details of the image format

```
(kali㉿kali)-[~/usb]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 1474560
Sector size: 512
```

- Show file system details and statistics including layout, sizes, and labels.

```
(kali㉿kali)-[~/usb]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3MM|
```

- List allocated and deleted file names in a directory

```
(kali㉿kali)-[~/usb]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(kali㉿kali)-[~/usb]
$
```

-d Display deleted entries only

```
(kali㉿kali)-[~/usb]
$ fls -d Ch01InChap01.dd
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

- Recover deleted file using inode

```
(kali㉿kali)-[~/usb]
$ icat Ch01InChap01.dd 15 > letter1.txt

(kali㉿kali)-[~/usb]
$ cat letter1.txt
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
```

- Extract the data units of letter1.txt (inode15)

```
(kali㉿kali)-[~/usb]
$ icat Ch01InChap01.dd 15
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
```

- Display the statistics and details about a given meta data structure

```

(kali㉿kali)-[~/usb]
$ istat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:09 (EST)

Sectors:
312

```

- List the details about data units and can extract the unallocated space of the file system

```

(kali㉿kali)-[~/usb]
$ blkls Ch01InChap01.dd 312-312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

```

- Extract the contents of a given data unit in Hex

```

(kali㉿kali)-[~/usb]
$ blkcat Ch01InChap01.dd 312 | xxd
00000000: 4561 726c 2c20 0d0a 5765 206e 6565 6420 Earl, ..We need
00000010: 746f 206d 6565 7420 6f6e 2074 6865 2031 to meet on the 1
00000020: 3874 6820 6f66 2041 7567 7573 7420 746f 8th of August to
00000030: 2063 6f6e 6669 726d 2074 6865 2077 6f72 confirm the wor
00000040: 6b20 4920 616d 200d 0a64 6f69 6e67 2066 k I am ..doing f
00000050: 6f72 2079 6f75 2e20 506c 6561 7365 2063 or you. Please c
00000060: 6f6e 7461 6374 206d 6520 4153 4150 2e0d ontact me ASAP..
00000070: 0a0d 0a47 656f 7267 6500 0000 0000 0000 ...George.....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000100: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000110: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000120: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000130: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000140: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000150: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000160: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000170: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000180: 0000 0000 0000 0000 0000 0000 0000 0000 .....
00000190: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000 ...George.....
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001c0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000 .....
000001f0: 0000 0000 0000 0000 0000 0000 0000 0000 .....

```

- Show A file with multiple sectors

```

(kali㉿kali)-[~/usb]
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

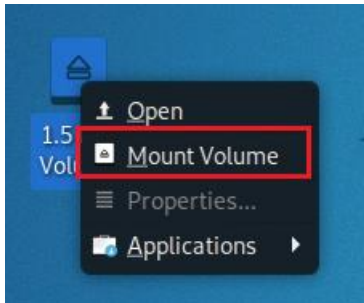
```

1. Assignment

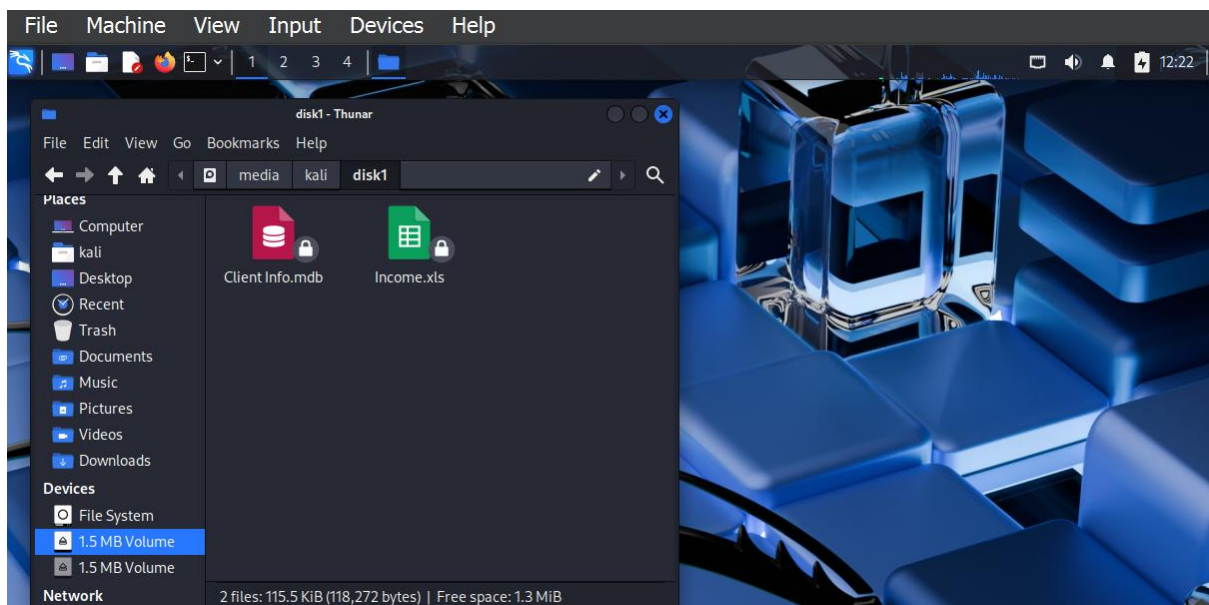
Mounting a USB image

```
(kali㉿kali)-[~/usb]
$ sudo losetup --partscan --find --show --read-only Ch01InChap01.dd
[sudo] password for kali:
/dev/loop0
```

- Mount the USB image



- Show content



- Show the disk content

```
(kali㉿kali)-[~/usb]
$ cd /media/kali/disk1

(kali㉿kali)-[/media/kali/disk1]
$ ls -l
total 116
-rw-r--r-- 1 kali kali 104448 Dec  9  2005 'Client Info.mdb'
-rw-r--r-- 1 kali kali  13824 Dec  9  2005  Income.xls

(kali㉿kali)-[/media/kali/disk1]
$
```

Exercise

1. Follow the DOCX and practice all Linux commands
2. How many deleted files in the disk image?
3. Recover all deleted files
 - Without mounting
 - Only using the commands introduced in this lecture

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 13.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 13 From YOUR NAME**", replacing "YOUR NAME" with your real name.